



LEARNING & DEVELOPMENT SYSTEM ARCHITECTURE

HRMS Capstone · Learning & Development Module

PHP 8.2.12 · MariaDB 10.4.32
502 PHP files · 57 1d_ tables
9 integration endpoints

PRESENTATION LAYER

Web Browser / Client



HTML5 CSS3 JavaScript

No front-end framework · custom design-token CSS

★ Font Awesome 6.5.1
🍃 Quill Editor 1.3.7
📦 Vanilla JS modules components · layout · utils

Responsive desktop · tablet · mobile sidebar, realtime notification polling

APPLICATION LAYER

Server-Side (Business Logic)



Core Modules / Features



Moderation Grade Book Knowledge Transfer Archive
Audit Log Admin Settings

Role portals: Administrator · Instructor / Trainer · Learner — routed by classes/Page.php

SYSTEM LIBRARIES / SERVICES

Domain classes & shared services

🔑 API Key Authentication classes/apiauth.php
📄 Integration Event Logger classes/integrationlog.php
🔒 CSRF Guard classes/csrf.php
📁 File Upload Service Dropzone component
📅 Certificate Engine & Expiry Checker
🎯 Skill & Competency Engine
📊 Analytics Aggregation
🕒 Snapshot Jobs skill & engagement

35 domain classes in classes/ — course, module, lesson, quiz, enrollment, progress, certificate, skill, program, learningpath, videoconference, grade...

INTEGRATION LAYER

REST API · JSON · X-API-Key

INBOUND — RECEIVE
👤 Employee Profile ← Employee Records
📊 Appraisal Data ← Performance
✅ Job Test Result ← Recruitment
★ Recognition Eligibility ← Rewards

OUTBOUND — PUBLISH
👤 Training Completion → pm_employee_training
📊 Learning Performance → Performance
🛡️ Compliance Training Log → lc_trainings
🕒 Attendance Data → ta_attendance
🎯 Workforce Analytics → wfa_*

Auth: X-API-Key / Bearer · scope learning-development
Tables: ld_api_key · ld_integration_log · ld_integration_event

DATA LAYER

Database (Storage)



MariaDB 10.4.32
database hrms

CONTENT HIERARCHY
ld_course · ld_module · ld_lesson · ld_quiz · ld_quiz_question · ld_quiz_question_option

ENROLLMENT & PROGRESS
ld_enrollment · ld_progress · ld_quiz_attempt · ld_quiz_session · ld_grade · ld_certificate · ld_evaluation_feedback

PATHS, PROGRAMS & TEMPLATES
ld_learning_path · ld_learning_path_item · ld_program · ld_course_template · ld_course_version · ld_prerequisite

SKILLS & COMPETENCY
ld_skill · ld_course_skill · ld_module_skill · ld_skill_snapshot · ld_engagement_snapshot · ld_training_recommendation

COLLABORATION & COMMUNICATION
ld_notification · ld_message · ld_comment · ld_note · ld_calendar_event · ld_announcement · ld_rating · ld_bookmark

INTEGRATION & GOVERNANCE
ld_api_key · ld_integration_log · ld_integration_event · ld_audit_log · ld_setting · ld_report

SHARED HRMS TABLES
em_employees · user_account · em_roles · em_departments



SECURITY

🔑 Authentication & Authorization
👤 Role-Based Access Control
📄 Session Management
📄 Data Validation & Sanitization
🔒 API Key Authentication
📋 Audit Logging

Dual sign-in: HRMS staff portal + L&D portal. Queries use **PDO prepared statements**.

INFRASTRUCTURE LAYER

Server Environment



XAMPP Stack

Apache PHP 8.2.12
MariaDB 10.4.32
phpMyAdmin 5.2.1

📁 Local File Storage assets/uploads
certificates · course thumbnails
lesson materials · lesson media
profile photos

Host: Windows server · deployment = copy + schema sync via ld_tables schema.sql

USERS

System End Users



System Administrator



L&D Administrator



Instructor / Trainer



Manager / Supervisor



Learner / Employee



Compliance Officer / Auditor